YENEPOYA (Deemed to be University)

SCHOOL OF SCIENCES

DEPARTMENT OF COMPUTER SCIENCE

A Mini Project Report on

RFID ATTENDANCE SYSTEM

In

Internet of Things

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1. About Internet of Things (IoT)

IoT refers to a system in which physical devices are connected to a network so that they can sense, process and exchange information. These devices generally contain sensors or identification modules, a controller, communication capability and software for processing data. IoT makes it possible to automate routine activities and collect useful information with minimal manual intervention.

The RFID Attendance System is an IoT-oriented application that combines RFID identification, an ESP32 microcontroller, a display and an internet connection. When a registered RFID card is presented to the reader, the system identifies the user, gives an immediate indication through the LCD/buzzer/LED and can send the attendance information to a cloud-based record.

2. About Technologies Used

Technology	Purpose
RFID	Contactless identification of a student using a unique RFID UID.
ESP32	Main controller; reads RFID data, controls output devices and provides Wi-Fi connectivity.
RC522	RFID reader module used to detect and read compatible RFID cards/tags.
I2C LCD	Displays attendance status and messages to the user.
Buzzer	Provides audible feedback for successful or unsuccessful scans.
LED	Provides visual status indication.
Wi-Fi	Provides network connectivity for cloud/data transfer.
Google Sheets / Cloud	Stores attendance records for later viewing and management.
Arduino IDE	Used to write, compile and upload the ESP32 program.

3. Abstract

Manual attendance systems require time, repeated data entry and physical handling of attendance registers. The RFID Attendance System is designed to automate this process using Radio Frequency Identification (RFID). Each student is assigned an RFID card or tag with a unique identifier. An RC522 RFID reader connected to an ESP32 detects the card and obtains

its UID. The ESP32 processes the UID and determines whether the card belongs to a registered student. The result is shown on an I2C LCD and is also indicated using a buzzer and LED.

With Wi-Fi connectivity, the system can transmit attendance information to a cloud-based spreadsheet or database. This reduces manual work, improves consistency and makes attendance records easier to maintain. The proposed system demonstrates how IoT technologies can be applied to a common educational requirement.

4. Objectives

- To design an automated attendance system using RFID technology.
- To identify students using the unique UID of an RFID card/tag.
- To use ESP32 as the central controller for RFID reading and output control.
- To provide immediate attendance feedback through an LCD, buzzer and LED.
- To use Wi-Fi connectivity for transferring attendance information to a cloud platform.
- To reduce manual attendance entry and improve the speed of attendance collection.
- To maintain attendance information in a digital format that can be reviewed later.

5. About the Project

The RFID Attendance System is intended for classrooms, laboratories and other environments where attendance must be recorded regularly. The system consists of an RFID reader connected to an ESP32. Students scan their RFID cards near the reader. The UID received from the card is compared with the registered student information. For a valid UID, attendance is recorded and a confirmation is displayed. For an unregistered UID, the system displays an appropriate message and provides an alert.

Basic working sequence:

- Power on the ESP32 and connected modules.
- Initialize the RFID reader, LCD, buzzer and LED.
- Connect the ESP32 to the configured Wi-Fi network.
- Wait for an RFID card/tag to be presented.
- Read the unique RFID UID.
- Compare the UID with registered student records.
- Display the student/attendance status and provide audio/visual feedback.
- Transmit or store the attendance record in the cloud when connectivity is available.
- Return to the scanning state for the next student.

6. System Requirements

6.1 Hardware Requirements

Component	Requirement / Role
ESP32 Development Board	Main IoT controller and Wi-Fi communication
RC522 RFID Module	Reads RFID cards/tags
RFID Cards/Tags	Unique student identification
20×4 I2C LCD	Displays messages and attendance status
Buzzer	Audio indication
LED	Visual indication
Jumper Wires / Breadboard	Temporary circuit connections
USB Cable / Power Supply	Programming and powering the ESP32
Wi-Fi Network	Internet connectivity for cloud updates

6.2 Software Requirements

- Arduino IDE
- ESP32 board support package
- RFID/RC522 library
- I2C LCD library
- Wi-Fi/network libraries suitable for ESP32
- Cloud/spreadsheet communication method used by the final implementation
- Web browser for viewing cloud attendance records

7. System Design

The system follows a simple IoT architecture. The RFID card is the identification source, the RC522 is the reader, the ESP32 performs processing, the LCD/buzzer/LED provide local feedback, and Wi-Fi provides the link to the cloud record.

System Block Diagram – RFID Attendance System

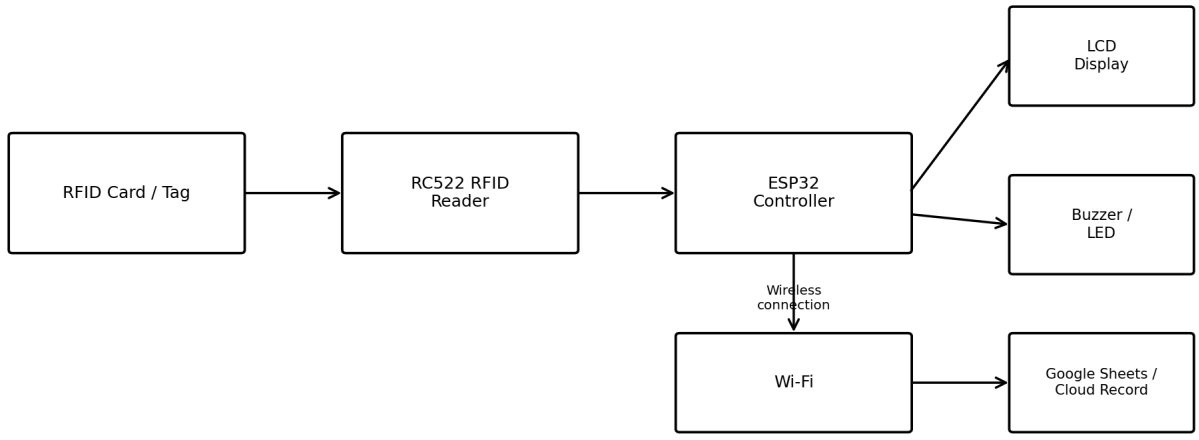


Figure 1: Block diagram of the RFID Attendance System

7.1 System Flow

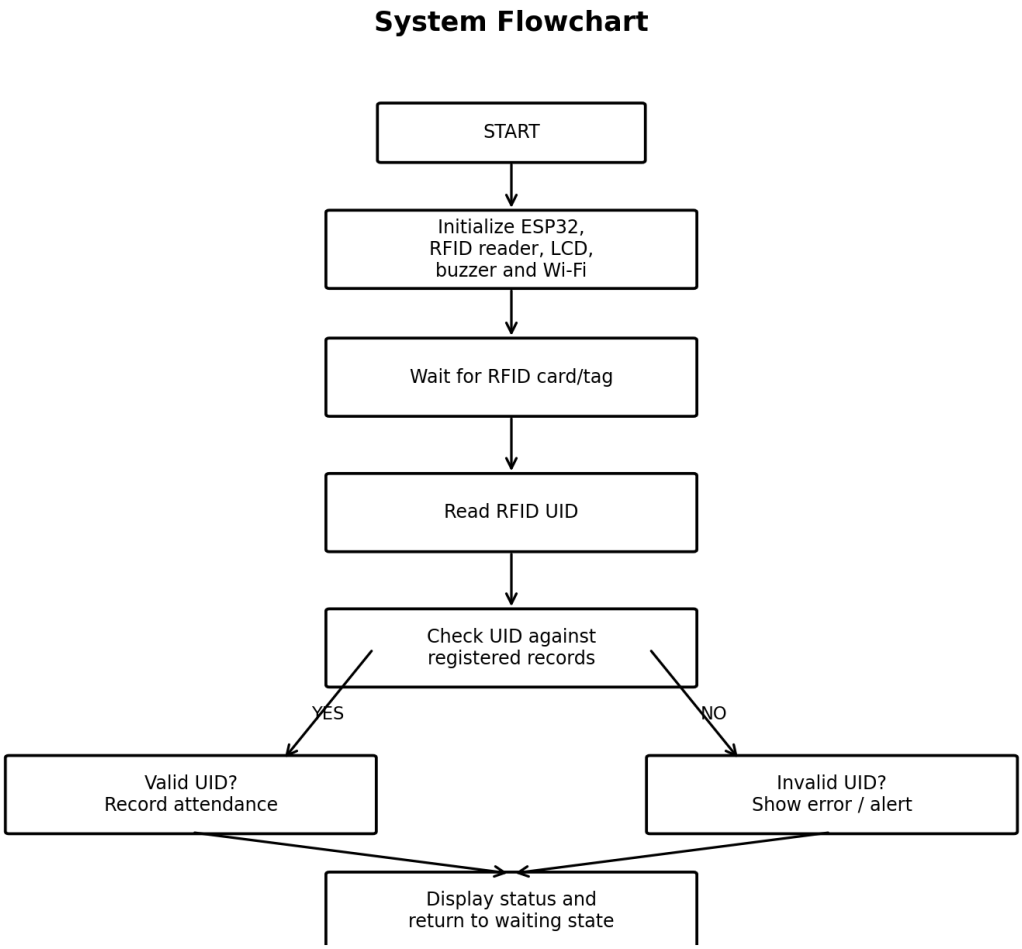


Figure 2: Flowchart of the attendance process

7.2 Working Principle

RFID communication allows the reader to obtain the identifier stored on a compatible RFID card/tag without requiring physical contact. The RC522 passes the UID to the ESP32. The ESP32 uses programmed registration logic to determine the associated student. After validation, the system updates the display and status indicators. If the system is configured for cloud recording, the ESP32 sends the relevant attendance data through Wi-Fi.

8. Hardware and Software Components

8.1 ESP32

ESP32 is a microcontroller platform with integrated Wi-Fi and Bluetooth capabilities. In this project it acts as the central controller. It receives RFID data, executes the attendance logic, controls the LCD, buzzer and LED, and handles network communication.

8.2 RC522 RFID Reader

The RC522 is an RFID reader/writer module commonly used with 13.56 MHz compatible RFID cards and tags. It communicates with the controller through a digital interface and provides the card UID needed for identification.

8.3 RFID Card/Tag

Each card/tag provides a unique identifier. The project uses this identifier as the student's digital identity for attendance purposes. A registration stage is used to associate a UID with a student.

8.4 I2C LCD

The I2C LCD provides local text output while using a small number of controller pins. It can show instructions such as 'Scan Card', successful attendance messages and invalid-card notifications.

8.5 Buzzer and LED

The buzzer and LED provide immediate feedback. A successful scan can be represented by a confirmation tone and LED indication, while an invalid or unsuccessful scan can use a different alert pattern.

8.6 Wi-Fi and Cloud Record

Wi-Fi allows the ESP32 to communicate with an online service. Attendance entries can include student identity, date/time and attendance status. A cloud spreadsheet such as Google Sheets can be used for simple record keeping.

9. Implementation

9.1 Hardware Setup

The modules are connected to the ESP32 according to their communication interfaces. The RC522 uses its digital RFID interface, while the I2C LCD uses SDA and SCL. The buzzer and LED are connected to suitable digital output pins. All modules share a common ground and are powered according to their electrical requirements.

Because ESP32 pin assignments can vary between development boards and project builds, the final wiring diagram should document the exact GPIO pins used in the assembled prototype. The connections must match the GPIO definitions in the final Arduino program.

9.2 Software Implementation

- Install the required ESP32 board package and libraries in Arduino IDE.
- Define the RFID, LCD, buzzer and LED interfaces.
- Initialize Serial communication for testing and debugging.
- Initialize the RFID reader and LCD.
- Connect the ESP32 to Wi-Fi.
- Continuously check for a new RFID card.
- Read and format the UID.
- Compare the UID with registered student data.
- Record attendance and provide feedback for a valid student.
- Show an error/alert for an invalid or unregistered card.
- Send the attendance information to the configured cloud platform when required.

10. Program Code

The following is a documentation reference implementation showing the core logic. The GPIO values and cloud request section must match the final hardware wiring and cloud endpoint used in the submitted prototype.

```
#include <SPI.h>
#include <MFRC522.h>
#include <Wire.h>
#include <LiquidCrystal_I2C.h>
#include <WiFi.h>

// Replace these with the GPIO pins used in the final wiring.
#define SS_PIN   5
#define RST_PIN  22
#define BUZZER   25
#define LED_PIN  26

MFRC522 rfid(SS_PIN, RST_PIN);
LiquidCrystal_I2C lcd(0x27, 20, 4);

const char* WIFI_SSID = "YOUR_WIFI_NAME";
const char* WIFI_PASS = "YOUR_WIFI_PASSWORD";

struct Student {
  String uid;
  String name;
};

Student students[] = {
  {"UID_1", "Student 1"},
  {"UID_2", "Student 2"},
  {"UID_3", "Student 3"}
};

void showMessage(String line1, String line2 = "") {
  lcd.clear();
  lcd.setCursor(0, 0);
  lcd.print(line1);
  lcd.setCursor(0, 1);
  lcd.print(line2);
}

int findStudent(String uid) {
  for (int i = 0; i < (int)(sizeof(students) / sizeof(students[0])); i++) {
    if (students[i].uid == uid) return i;
  }
  return -1;
}

String readUID() {
  String uid = "";
  for (byte i = 0; i < rfid.uid.size; i++) {
    if (rfid.uid.uidByte[i] < 0x10) uid += "0";
    uid += String(rfid.uid.uidByte[i], HEX);
    if (i < rfid.uid.size - 1) uid += ":";
  }
  uid.toUpperCase();
  return uid;
```

```
}

void setup() {
  Serial.begin(115200);

  pinMode(BUZZER, OUTPUT);
  pinMode(LED_PIN, OUTPUT);
  digitalWrite(BUZZER, LOW);
  digitalWrite(LED_PIN, LOW);

  SPI.begin();
  rfid.PCD_Init();

  lcd.init();
  lcd.backlight();
  showMessage("RFID Attendance", "Scan your card");

  WiFi.begin(WIFI_SSID, WIFI_PASS);
  while (WiFi.status() != WL_CONNECTED) {
    delay(500);
    Serial.print(".");
  }

  Serial.println("\nWi-Fi connected");
  showMessage("Wi-Fi Connected", "Scan your card");
  delay(1000);
}

void loop() {
  if (!rfid.PICC_IsNewCardPresent()) return;
  if (!rfid.PICC_ReadCardSerial()) return;

  String uid = readUID();
  Serial.println("UID: " + uid);

  int index = findStudent(uid);

  if (index >= 0) {
    digitalWrite(LED_PIN, HIGH);
    tone(BUZZER, 2000, 150);

    showMessage("Attendance Marked", students[index].name);
    // Add the project-specific cloud request here.

    delay(2000);
    digitalWrite(LED_PIN, LOW);
  } else {
    tone(BUZZER, 500, 500);
    showMessage("Invalid Card", "Not Registered");
    delay(2000);
  }

  rfid.PICC_HaltA();
  rfid.PCD_StopCrypto1();

  showMessage("RFID Attendance", "Scan your card");
}
```

11. Database / Cloud Platform

For digital attendance management, the project can use Google Sheets as a lightweight cloud-based record. The ESP32 sends the required attendance information through the project's configured web/API method. A typical record contains the student's name or identifier, RFID UID, date, time and attendance status.

Field	Description
Student Name	Name associated with the registered RFID UID
RFID UID	Unique identifier read from the card/tag
Date	Date on which attendance was recorded
Time	Time at which the card was scanned
Status	Attendance result such as Present

The cloud record provides a centralized digital history that can be reviewed by authorized users. The exact communication endpoint, authentication method and spreadsheet columns should correspond to the final deployed configuration.

12. Output / Screenshots

This section should contain photographs/screenshots from the completed prototype. Recommended evidence includes:

- Complete hardware prototype showing ESP32, RC522, LCD, buzzer and LED.
- LCD showing the initial 'Scan your card' message.
- Successful RFID scan showing attendance confirmation.
- Invalid/unregistered card alert.
- Serial Monitor output showing the detected UID.
- Cloud spreadsheet showing the recorded attendance entry.

Figure 1: Initial RFID Attendance System ready for card scanning.

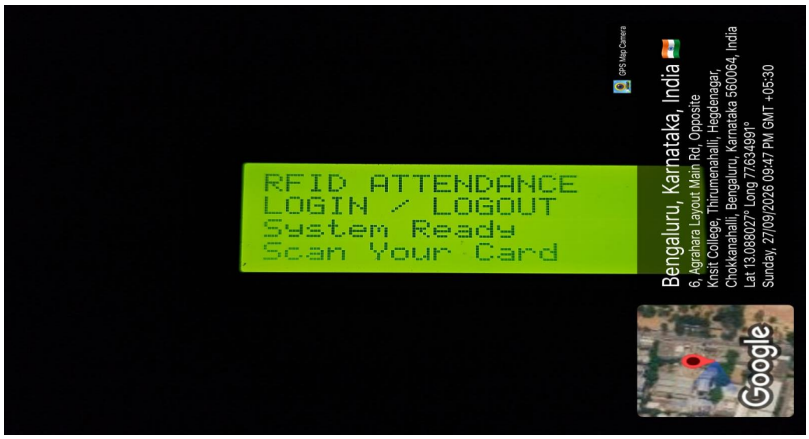


Figure 2: RFID login/welcome message displayed for a registered user.

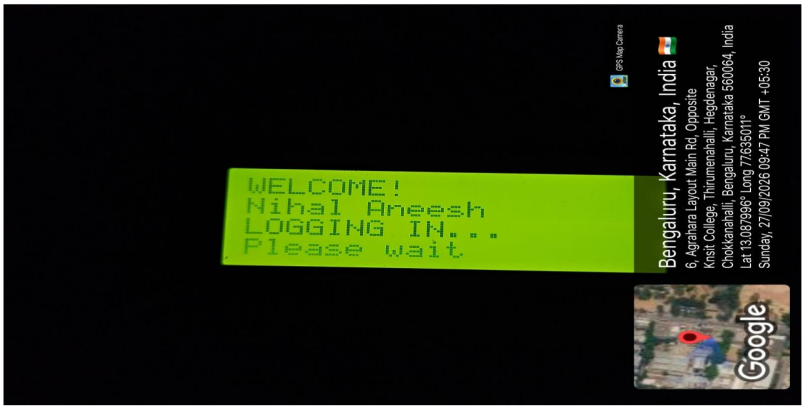
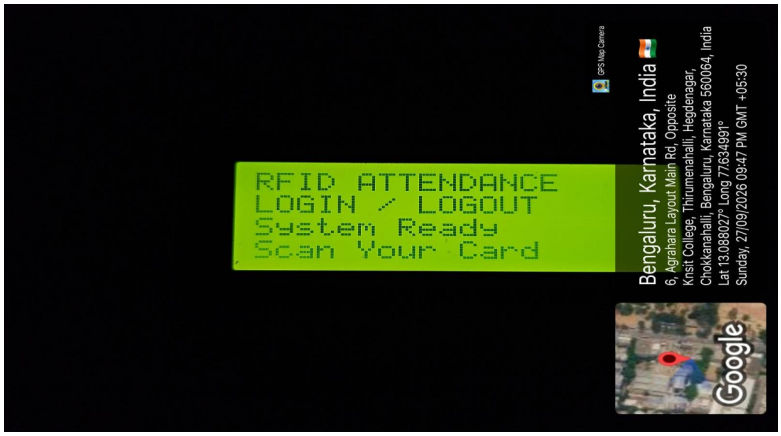


Figure 3: RFID logout message displayed after the user logs out.



13. Results and Discussion

The implemented design demonstrates the feasibility of automated RFID-based attendance recording. RFID provides a quick identification mechanism, while the ESP32 combines identification, local feedback and network connectivity in one controller. The LCD makes the system easy to operate because users receive immediate instructions and results.

Test Case	Expected Result	Observed/To Be Documented
Registered RFID card	Student identified and attendance marked	Record the actual successful output
Unregistered RFID card	Invalid-card message and alert	Record the actual alert output
No card present	System waits for a scan	Record standby display
Wi-Fi available	Cloud update can be attempted	Record successful cloud entry
Wi-Fi unavailable	Local identification can still be indicated; cloud update depends on implementation	Record final prototype behaviour

The project reduces the repetitive work associated with manual attendance and provides a digital record. Reliability depends on correct UID registration, stable power, proper RFID-reader wiring and network availability when cloud synchronization is used.

14. Conclusion

The RFID Attendance System provides an automated approach to recording attendance using RFID and IoT technology. The RC522 identifies the RFID card, the ESP32 processes the UID and controls the output devices, and Wi-Fi enables the attendance data to be connected to a cloud record. The LCD, buzzer and LED provide clear local feedback to the user.

The project demonstrates the practical use of IoT concepts in an educational environment and provides a foundation for building a more advanced, scalable attendance-management solution.

15. Future Enhancement

- Add a real-time web dashboard for teachers or administrators.
- Introduce secure user authentication for administrators.
- Prevent duplicate attendance within the same session.
- Add offline data buffering and automatic synchronization after Wi-Fi recovery.
- Add date/time synchronization using an NTP server or a real-time clock.
- Support multiple classrooms and centralized attendance reports.
- Add export options for monthly or semester attendance.
- Improve enclosure design and power management for a permanent installation.
- Add additional identification methods such as QR or biometric verification where appropriate.

16. References

1. Arduino Documentation – Arduino IDE and microcontroller programming resources.
2. Espressif Systems – ESP32 documentation and development resources.
3. MFRC522 / RFID library documentation and module reference material.
4. LiquidCrystal_I2C library documentation.
5. Google Sheets documentation and spreadsheet-based data management resources.
6. Course materials and laboratory guidance for Internet of Things.